



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

INTRODUCTION AUX MÉTA-ANALYSES ET REVUES SYSTÉMATIQUES

Planning a Systematic Review: From Formulating the Research Questions to Developing and Publishing the Protocol

Joseph LANGRIDGE – Chargé de mission scientifique «
Revue systématique et synthèses »

SOUTIENS INSTITUTIONNELS



PLAN

1

Needs and question types

2

Formulating *review* questions

3

Importance of a Protocol

4

Problems and solutions

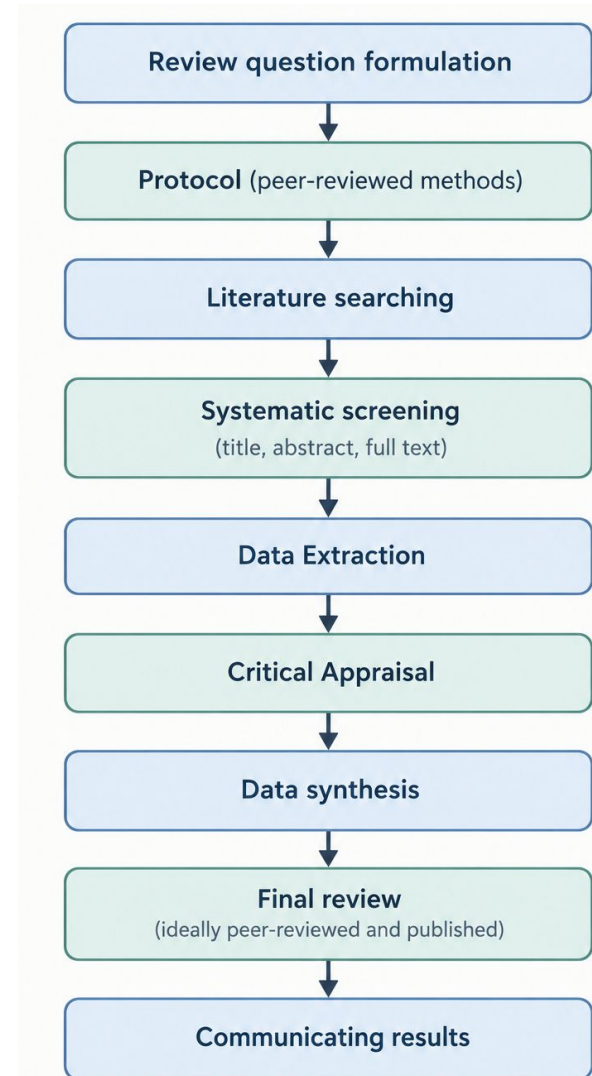
5

How and where to publish the protocol

Planning a systematic review

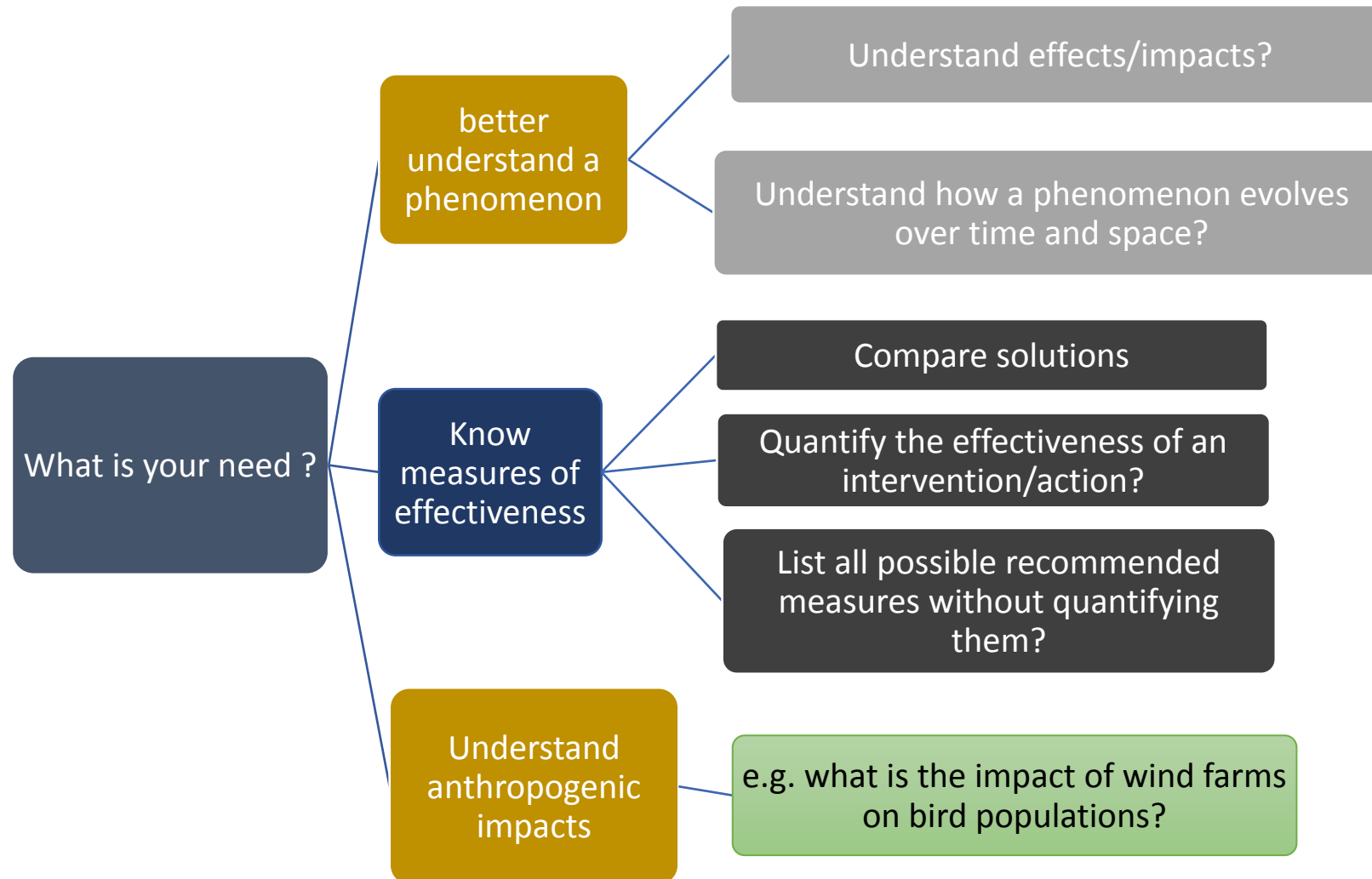
Most importantly – at the beginning of the process!

... don't wing it!



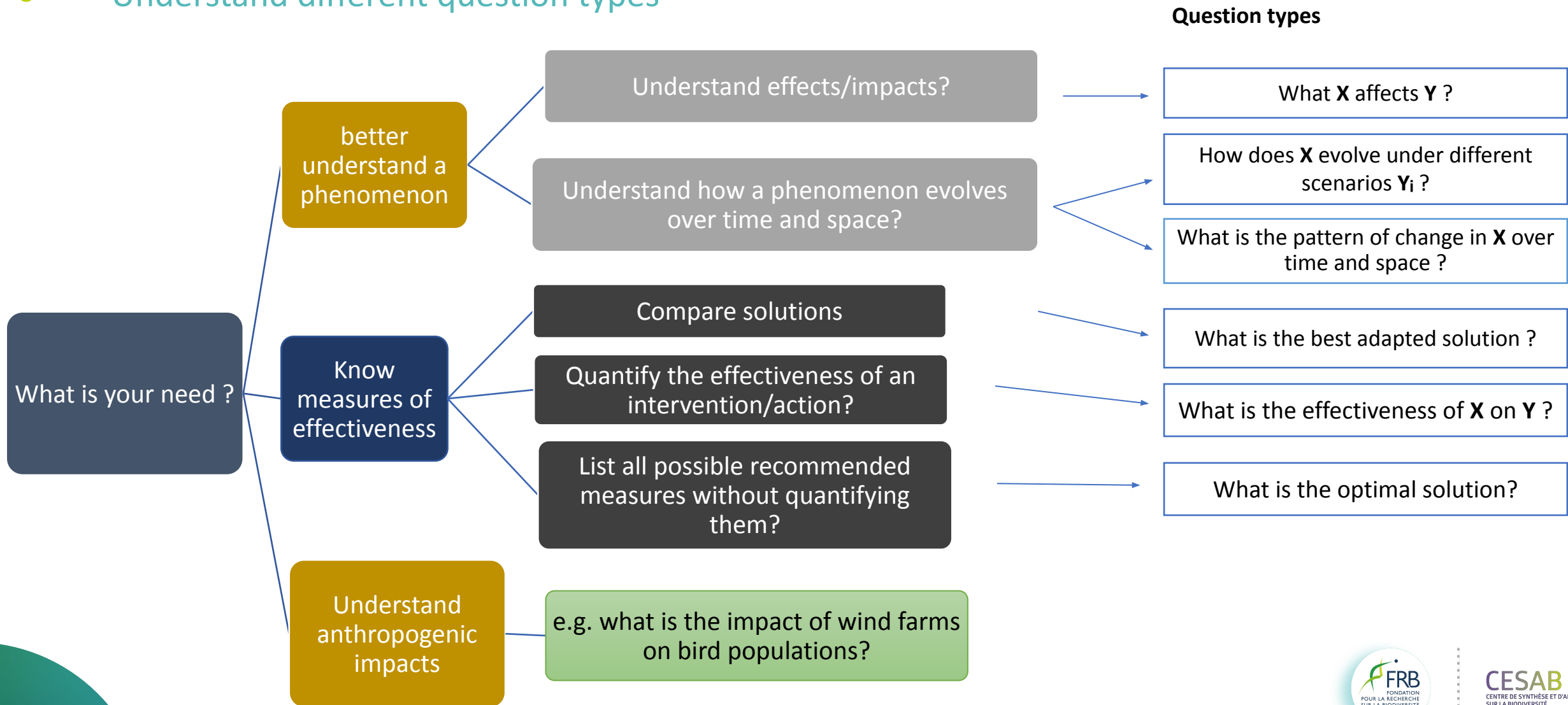
Needs and question types

Understand a stakeholders' needs



Needs and question types

Understand different question types



Formulating review questions

Invest time in planning

- Identify the question that matters **most to key stakeholders and policymakers**.
- **Maximise cost-effectiveness** by making the most efficient use of available time and resources.
- **Minimise confusion** by avoiding vague or inappropriate wording.

Guidelines and Standards for Evidence Synthesis in Environmental Management



Section 2

Identifying the need for evidence, determining the Evidence Synthesis type, and establishing a Review Team

<https://environmentalevidence.org/information-for-authors/2-need-for-evidence-synthesis-type-and-review-team/>



Formulating review questions

A project pipeline analogy

- Invest time upfront to design a clear, straight pipeline
- The goal is to keep the **flow of the project smooth, efficient, and directed toward the outcomes that matter most.**



Formulating review questions

A project pipeline analogy

Avoid unnecessary **bends, ups and downs, or bottlenecks**



Formulating review questions

Invest time in planning

- Identify and define the different components or concepts (e.g. use question frameworks i.e., PICO, PECO...)
- Determine which articles best address the chosen concepts (e.g., undertake a *pre-scoping*)

Guidelines and Standards for Evidence Synthesis in Environmental Management

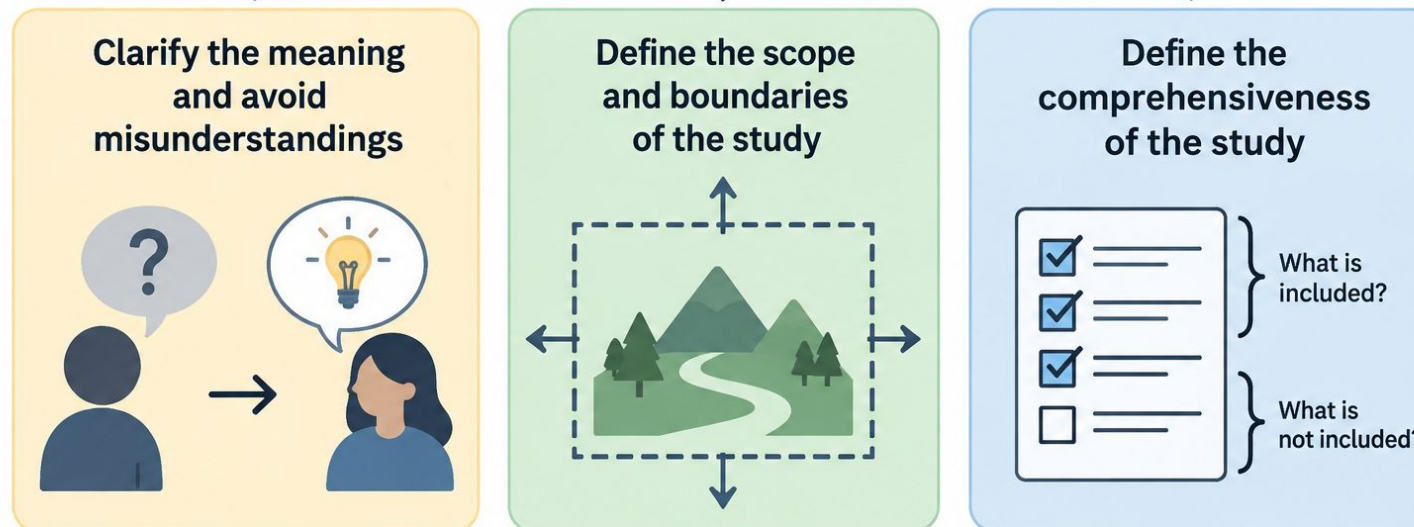


Formulating review questions

Invest time in planning

Formulating the question helps to:

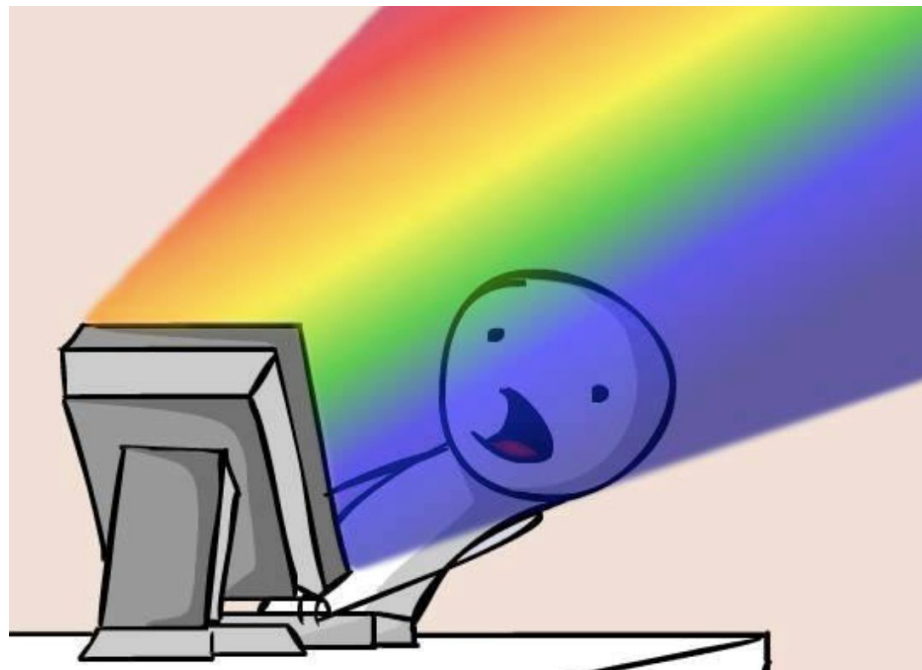
- Clarify meaning and semantics and avoid misunderstandings
- Define the scope and boundaries of the study
- Define the comprehensiveness of the study



Formulating review questions

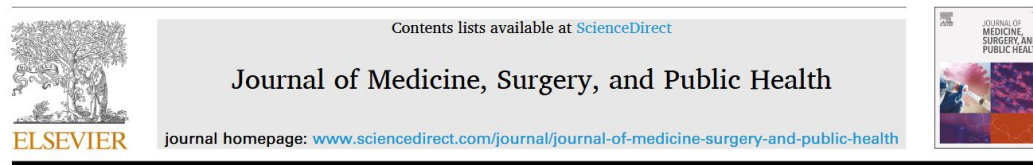
For example, some subjects can be a « semantic nightmare »

Actions to protect, **conserve**, **restore**, **sustainably use** and **manage** **natural** or **modified** **terrestrial**, **freshwater**, **coastal** and **marine** ecosystems, which address **social**, **economic** and **environmental** challenges *effectively* and *adaptively*, while simultaneously providing human **well-being**, **ecosystem services** and **resilience** and **biodiversity benefits**, UNEA-5



Formulating review questions

Identify different components and concepts



Formulating research questions for evidence-based studies

Mohammad-Salar Hosseini ^{*,1}

Table 1

Common approaches in formulating research questions, according to the type of study.

Type of study	Question framework	Components of the model
Interventional/ Effectiveness	PICO(T/S)	Patient/Population, Intervention, Comparator, Outcome, (Time/Study design)
Etiologic/Risk/Cause	PEO	Population, Exposure/Environment, Outcome
Incidence/Prevalence	CoCoPop	Condition, Context, Population
Diagnostic test accuracy	PIRD	Population, Index Test, Reference Test, Diagnosis of Interest
Prognostic	PFO	Population, Prognostic Factor (or model of interest), Outcome
Economic/Cost-effectiveness	PICOC	Population, Intervention, Comparator, Outcomes, Context
Methodologic	SDMO	Study type, Data type, Method type, Outcome
Theory/Model	BeHEMTh	Behavior of interest, Health context, Exclusions, Models or Theories
Psychometric	CPTM	Construct of interest (measurement instrument), Population, Type of measurement instrument, Measurement properties
Qualitative	PICo	Population, Phenomena of Interest, Context
Qualitative	PPhTS	Participants, Central Phenomenon, Time, and Space
Qualitative: Public health	SPICE	Setting, Perspective, Intervention, Comparison, Evaluation
Qualitative/Mixed-Method: Experience and views	SPIDER	Sample, Phenomenon of Interest, Design, Evaluation, Research type
Qualitative: Organizational and policy evaluation	CIMO	Context, Intervention, Mechanism, Outcome
Qualitative: Health policy and management	ECLIPSe	Expectation, Client group, Location, Impact, Professionals, Service
Scoping	PCC	Population, Concept, Context

Formulating review questions

PICO / PECO

Applying a PICO / PECO		
	...for an intervention approach	...for an exposition approach
Population (P)	<i>The study unit on which we measure the effect/impact of the intervention.</i>	<i>The study unit on which we measure the effect/impact of the exposure.</i>
Intervention ou Exposition (I/E)	The practiced <i>Intervention</i> itself having an effect on the population.	What the population is exposed to.
Comparator (C)	What is the effect of the intervention being compared to (control vs. intervention)?	What are we comparing the effect of the exposure (control) to?
Outcomes (O)	What indicator/metric is being measured in order to demonstrate an effect.	What indicator/metric is being measured in order to demonstrate an effect.
Context (C /T)	In what context (geographical and/or temporal)?	In what context (geographical and/or temporal)?

Formulating review questions

Example 1 : PICO



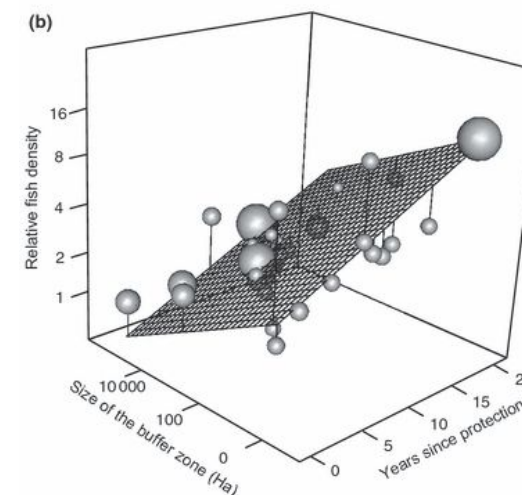
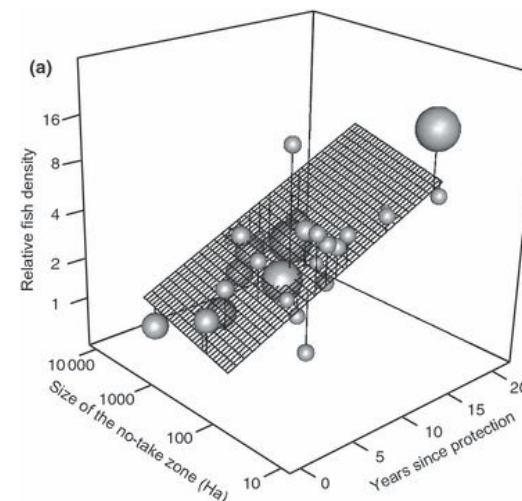
Full Access

Marine reserves: size and age do matter

Joachim Claudet✉, Craig W. Osenberg, Lisandro Benedetti-Cecchi, Paolo Domenici, José-Antonio García-Charton, Ángel Pérez-Ruzafa, Fabio Badalamenti, Just Bayle-Sempere, Alberto Brito, Fabio Bulleri, Jean-Michel Culioli, Mark Dimech, Jesús M. Falcón, Ivan Guala, Marco Milazzo, Julio Sánchez-Meca, Paul J. Somerfield, Ben Stobart, Frédéric Vandeperre, Carlos Valle, Serge Planes

Population	Intervention	Comparator (control)	Outcome (i.e. measurable indicators)
Commercial fish species	Established marine reserves (no-take zones)	No protection or before establishment of PA	Abundance, density

Claudet et al 2008, *Ecology Letters*. <https://doi.org/10.1111/j.1461-0248.2008.01166.x>



Formulating review questions

Example 2: PICO

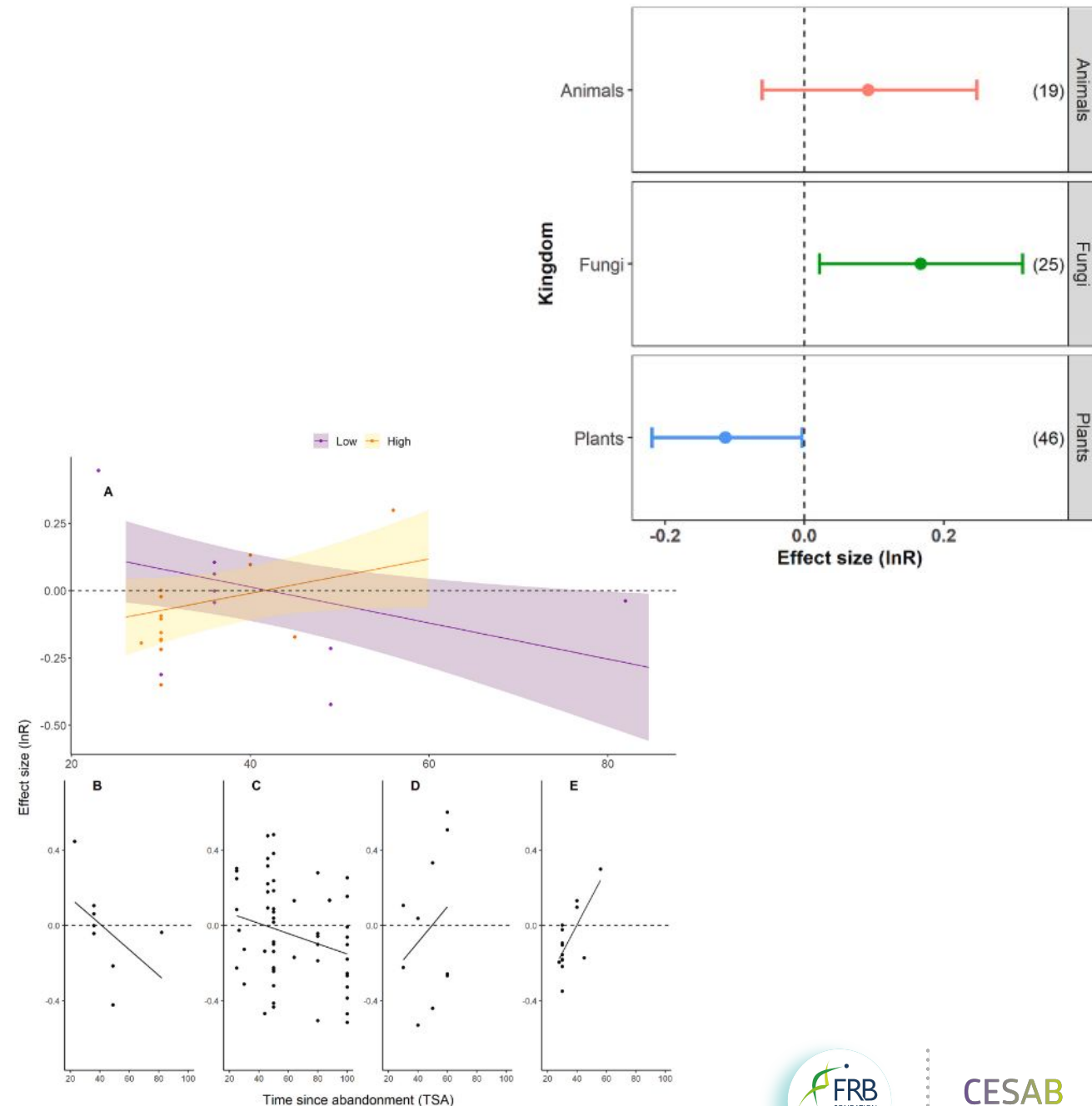


Review

Biodiversity responses to forest management abandonment in boreal and temperate forest ecosystems: A meta-analysis reveals an interactive effect of time since abandonment and climate

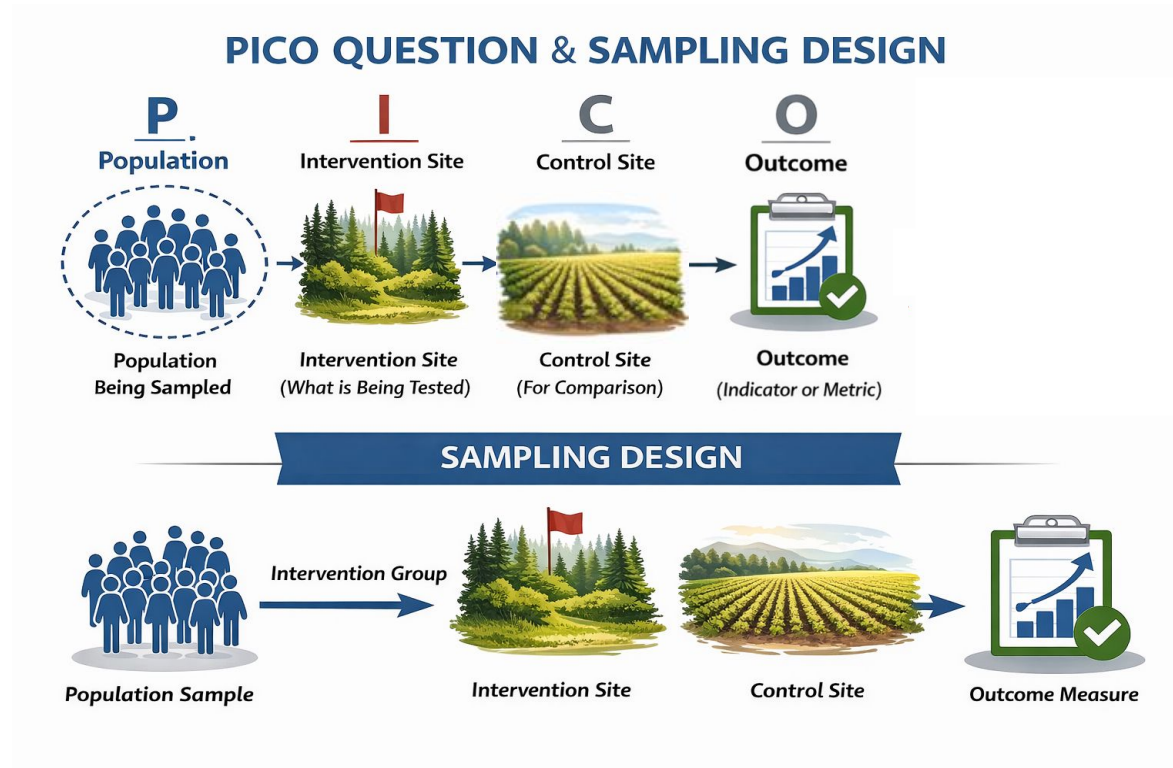
Joseph Langridge^{a,b,*}, Sylvain Delabye^{c,d,e}, Olivier Gilg^{f,g}, Yoan Paillet^h, Yorick Reyjol^a, Romain Sordello^a, Julien Touroult^a, Frédéric Gosselin^c

Population	Intervention	Comparator (control)	Outcome (i.e. measurable indicators)
Terrestrial biodiversity	Cessation of management (passive management)	Managed / exploited forests	Abundance, richness



Formulating review questions

The PICO and PECO are intrinsically linked to sampling designs



P for:

- Patients [age group, gender, etc.]
- **Population (species, taxa...)**
- Problem
- Participants
- Perspective

C for:

- Comparison
- Control group
- Control site
- Comparator

Formulating review questions

From broad to specific questions



Question	Key elements	Question type
What is the impact of roads on wildlife?	None are specified other than roads (vague), wildlife (vague) and impact (vague)	Open-framed (possible for Systematic Mapping but unsuitable for Systematic Review/meta-analysis)
What is the impact of motorways on populations of endemic bird species in Europe?	Motorways (=exposure), European endemic bird species (=population); but comparator and outcome not specified	Open-framed (suitable for Systematic Mapping but unsuitable for Systematic Review/meta-analysis)
What is the impact of habitats containing motorways on the breeding success of endemic European bird species, as compared to habitats without motorways?	Motorways (=exposure), no motorways (=comparator), European endemic bird species (=population), breeding success (=outcome)	Closed-framed (suitable for Systematic Mapping and possible for Systematic Review/meta-analysis)

LET'S GO!



Importance of a Protocol

So, now what ?

Let's write a protocol and get the methods peer-reviewed

Importance of a Protocol

What does the CEE say?

- A review protocol provides a **step-by-step guide** for conducting Evidence reviews.
- Develop an *a priori* protocol before starting the literature review so that **the process is clear and consistent**.
- The protocol should contain **specific guidelines to identify, screen relevant articles, extract data, and analyse the data**.
- The protocol can help other review teams **replicate the work or update a literature review** when new research becomes available.

Guidelines and Standards for Evidence Synthesis in Environmental Management



Section 4

Writing and registering a Protocol

<https://environmentalevidence.org/information-for-authors/4-writing-and-registering-a-protocol/>



Importance of a Protocol

Problems and solutions

Traditional approaches to reviewing literature may be susceptible to bias and result in incorrect decisions (Haddaway *et al.* 2020).

Despite the increasing popularity of systematic reviews in the environmental field, evidence synthesis methods continue to be poorly applied in practice (Haddaway *et al.* 2020).

PERSPECTIVE

<https://doi.org/10.1038/s41559-020-01295-x>

nature
ecology & evolution

 Check for updates

Eight problems with literature reviews and how to fix them

Neal R. Haddaway ^{1,2,3}✉, Alison Bethel⁴, Lynn V. Dicks^{5,6}, Julia Koricheva ⁷, Biljana Macura ²,
Gillian Petrokofsky⁸, Andrew S. Pullin⁹, Sini Savilaakso ^{10,11} and Gavin B. Stewart ¹²

Source: Haddaway et al. 2020. *Nature Ecology & Evolution*. <https://doi.org/10.1038/s41559-020-01295-x>



Importance of a Protocol

Problems and solutions

Scientific principles should be followed:

A protocol aims at objectifying the results/conclusions:

- **Replicability**, and
 - **Transparency**, through archiving/reporting
 - **Consideration of biases** (internal, external),
-
- ✓ provides a framework to ensure **reliability & credibility**
 - ✓ outlines a systematic approach

Maximizing reliability = published protocol + review (both peer-reviewed)



Importance of a Protocol

Problems and solutions

Mission creep:

Occurs when the review deviates from the initial objectives

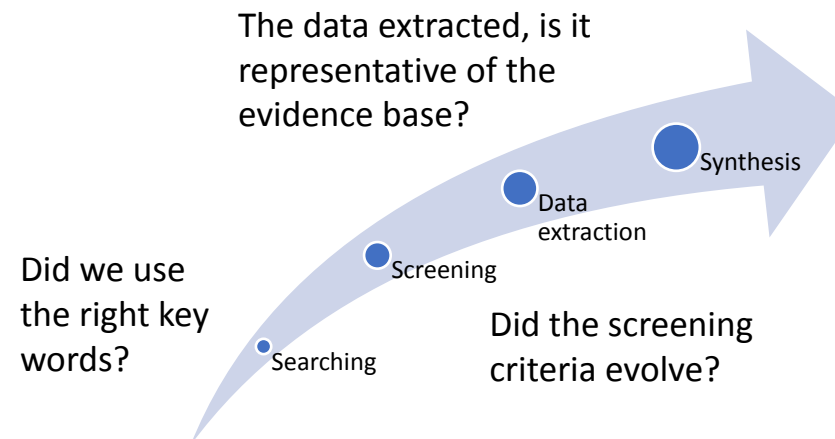
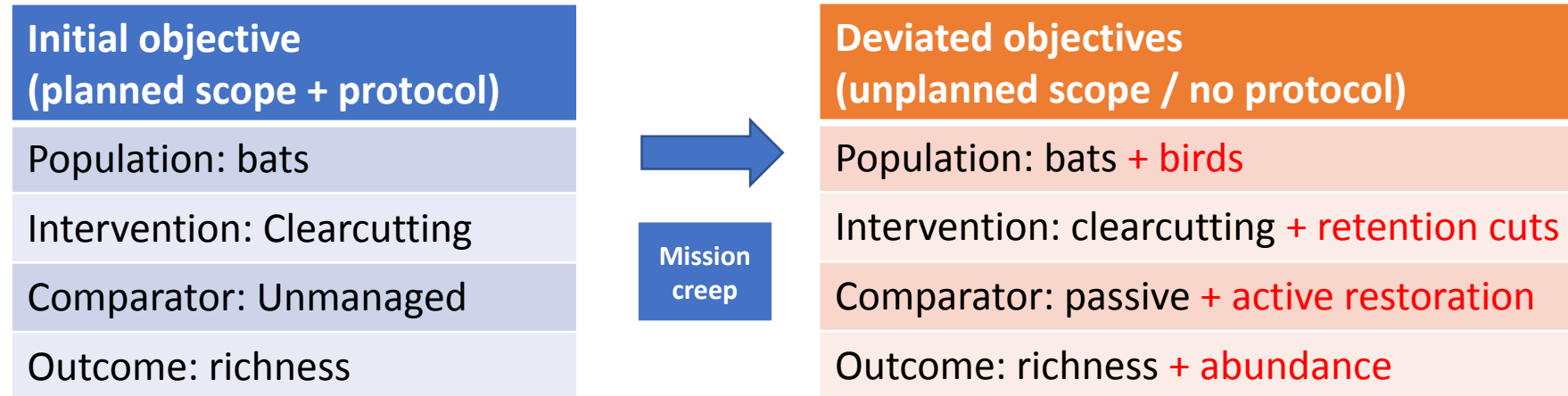
What elements can evolve during the process?

- Key definitions
- Search strategies and inclusion
- Appraisal criteria may alter over time or differ between reviewers

Source: Haddaway *et al.* 2020. Nature ecology and evolution. <https://doi.org/10.1038/s41559-020-01295-x>

Importance of a Protocol

Problems and solutions





Importance of a Protocol

Problems and solutions

Mission creep:

Occurs when the review deviates from the initial objectives

What are the consequences?

- **not representative** of the evidence base because **important studies may have been omitted**
- Inaccurate and misleading
- Unrepeatable, not upgradable, not updateable

Source: Haddaway *et al.* 2020. Nature ecology and evolution. <https://doi.org/10.1038/s41559-020-01295-x>



Importance of a Protocol

Lack of transparency/replicability:

An ability to repeat a review's methods exactly ('replicability')

If the reader can't understand:

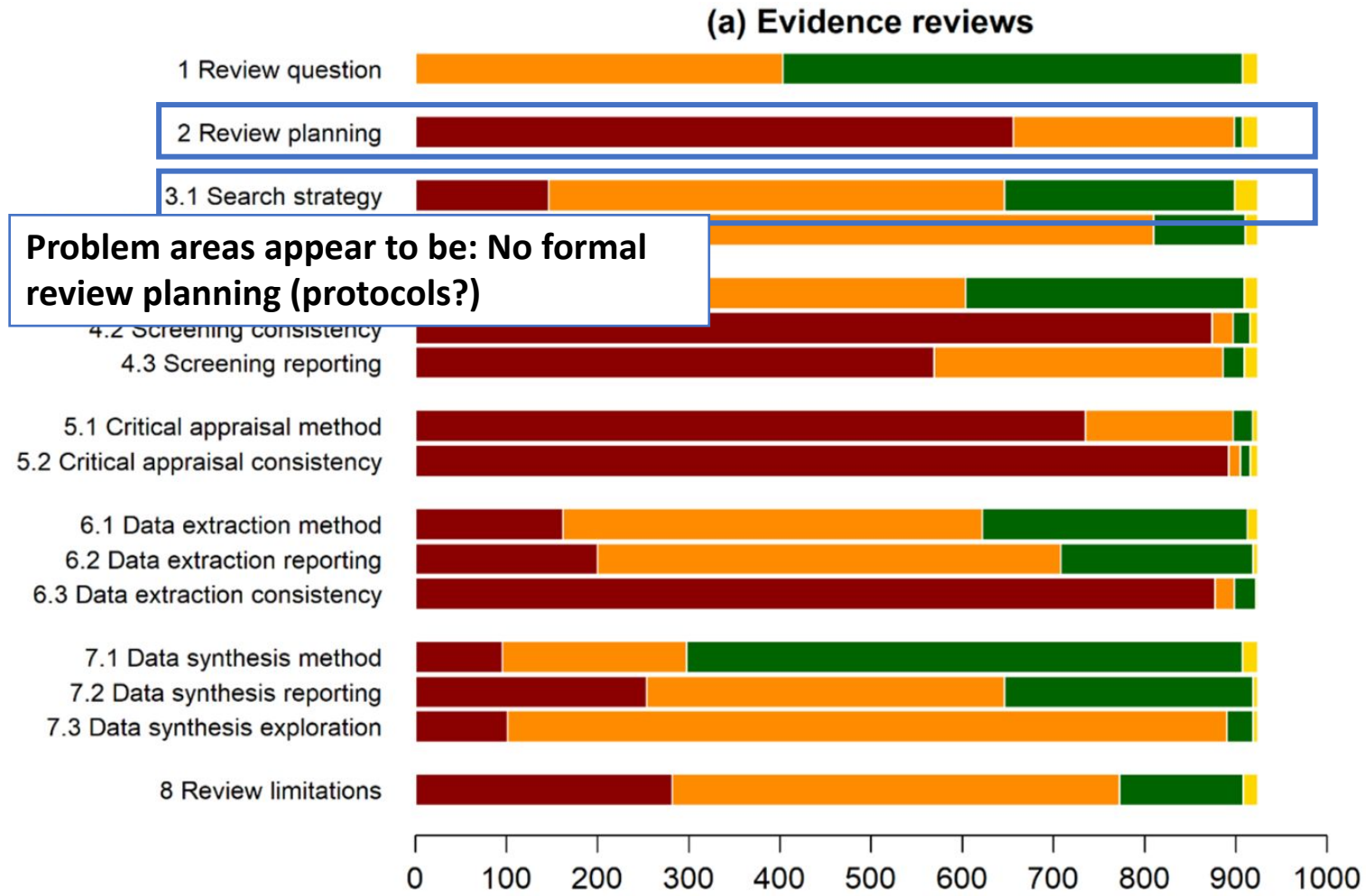
- *how studies were identified, selected and synthesized*
- *which ones were excluded,*

What are the consequences?

- Risk of bias cannot be assessed, and **unclear subjective decisions can't be fully trusted.**

Source: Haddaway *et al.* 2020. Nature ecology and evolution. <https://doi.org/10.1038/s41559-020-01295-x>

It calls into question the reliability and credibility



Source: Pullin *et al.* 2022. *Environmental Evidence*.
<https://doi.org/10.1186/s13750-022-00269-9>

It calls into question the reliability and credibility

- 92 reviews evaluated using CEESAT, published across **68 different peer-reviewed journals** and 3 grey literature sources

CEESAT criteria

1 Was an *a-priori* protocol available for comment before the synthesis was conducted?

2: Searching for studies

2.1 Does the search for literature utilise a comprehensive range of sources?

2.2 Are the search strings clearly defined?

3: Including studies

3.1 Does the synthesis apply clearly documented inclusion criteria to all potentially relevant studies found during the search?

3.2 Does the synthesis demonstrate that inclusion/exclusion decisions are repeatable?

3.3 Are inclusion/exclusion decisions transparent?

4: Critical Appraisal

4.1 Does the synthesis conduct and report critical appraisals of the methods of each study?

4.2 Are studies objectively weighted according to methodological quality?

5: Data Extraction

5.1 Is data extraction documented, repeatable and consistent?

5.2 Are the extracted data reported for each study?

6: Data synthesis

6.1 Is a quantitative synthesis conducted?

6.2 Is heterogeneity in the effect of the Intervention/Exposure investigated statistically?

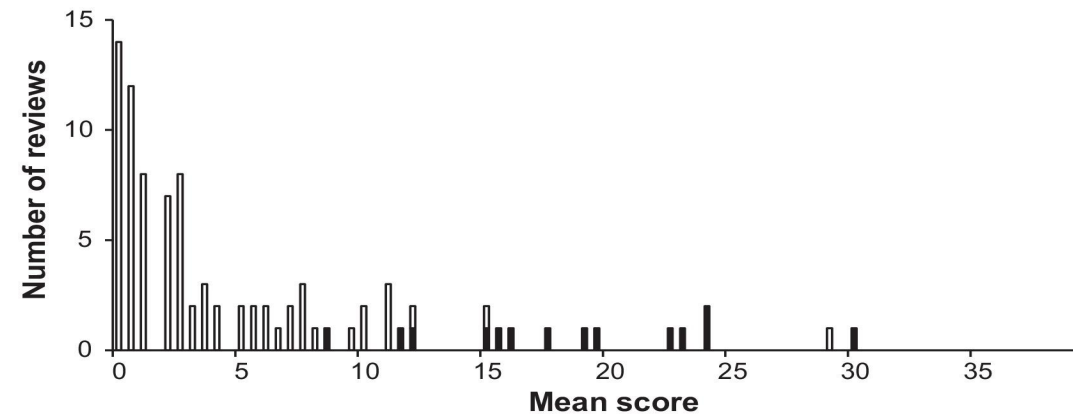
6.3 Does the synthesis consider possible publication bias?



The reliability of evidence review methodology in environmental science and conservation



Bethan C. O'Leary^{a,*}, Kristian Kvist^a, Helen R. Bayliss^a, Géraldine Derroire^b, John R. Healey^b, Kathryn Hughes^c, Fritz Kleinschroth^b, Marija Sciberras^c, Paul Woodcock^d, Andrew S. Pullin^a



Overall, the mean score was **5.8 (0-30/39)** but the median value was **2.5**

Source: O'Leary et al. 2016. *Environmental Science & Policy*. <https://doi.org/10.1016/j.envsci.2016.06.012>

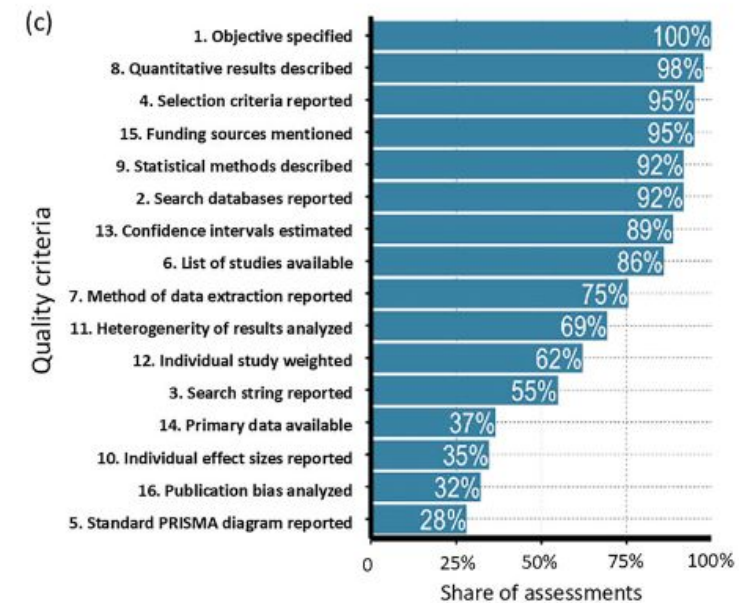
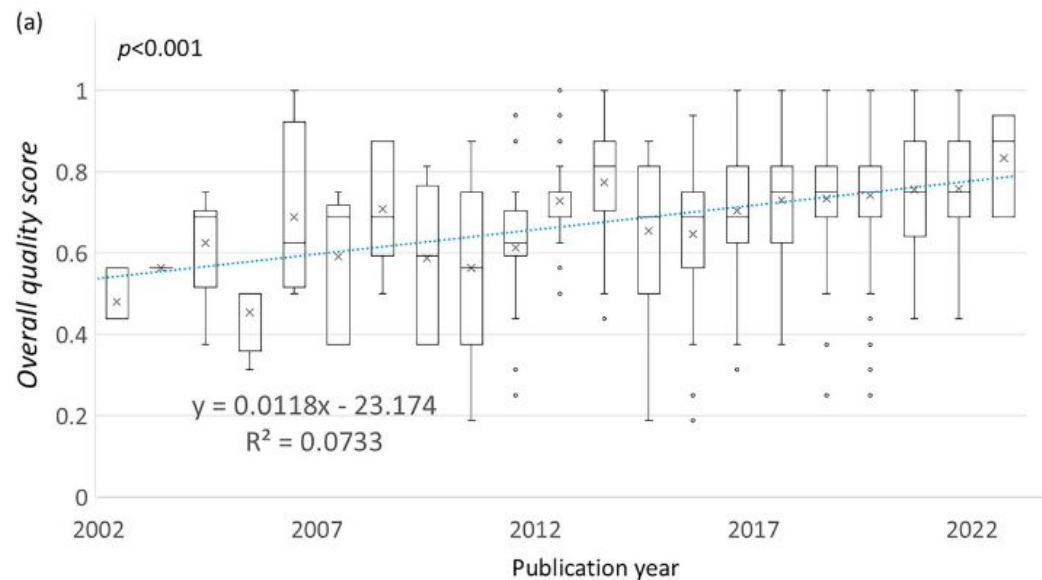


It calls into question the reliability and credibility

The quality of meta-analyses in agricultural sustainability has been increasing, but weaknesses persist

 Check for updates

Andrea Schievano¹✉, Simona Bosco¹, Marta Pérez-Soba², Ana Montero-Castaño¹, Rui Catarino¹, Mathilde Chen^{3,4,5}, Giovanni Tamburini⁶, Beatrice Landoni⁷, Otho Mantegazza¹, Carlo Rega¹, Irene Guerrero¹, Maria Bielza⁸, Marijn Van der Velde¹, Maria Luisa Paracchini¹, Linda See⁹, Jean-Michel Terres¹ & David Makowski³



Source: Schievano et al. 2026. *npj Sustain. Agric.* <https://doi.org/10.1038/s44264-026-00148-7>

Further reading: planning a systematic review

- Campbell Systematic Reviews: Policies and Guidelines (Campbell Collaboration, 2014). <https://onlinelibrary.wiley.com/pb-assets/assets/18911803/Campbell%20Policies%20and%20Guidelines%20v4-1559660867160.pdf>
- Higgins, J. P. et al. Cochrane Handbook for Systematic Reviews of Interventions (John Wiley & Sons, 2019). <https://training.cochrane.org/handbook>
- Shea, B. J. et al. AMSTAR 2: a critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both. BMJ 358, j4008 (2017). <https://www.bmj.com/content/358/bmj.j4008>
- Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) <https://www.prisma-statement.org/>
- RepOrting standards for Systematic Evidence Syntheses (ROSES) <https://www.roses-reporting.com/>

Guidelines and Standards for Evidence Synthesis in Environmental Management



Section 3

Planning a CEE Evidence Synthesis

<https://environmentalevidence.org/information-for-authors/3-planning-a-cee-evidence-synthesis/>



What should be included in a protocol?

1. Title

Type of evidence synthesis

2. Background / rationale

Why the review is needed

3. Outline objectives, Primary & secondary review question(s)

Use question frameworks e.g., PICO: population, intervention, outcomes

4. Methods

Study identification & synthesis process

a. Search strategy (**search terms**)

c. Article screening process (**Eligibility criteria**)

e. Study mapping methods

h. Data synthesis methods (**analysis**)

b. Bibliographic **sources** information

d. Data coding process (data **variables**, study **characteristics**)

f. Risk of bias assessment

Where can a protocol be published?

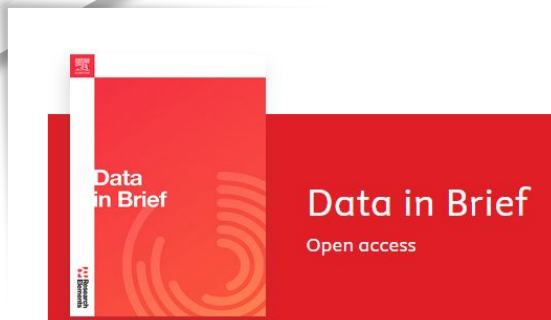


Editors: Marc Cadotte, Holly Jones, Carolyn Kurle

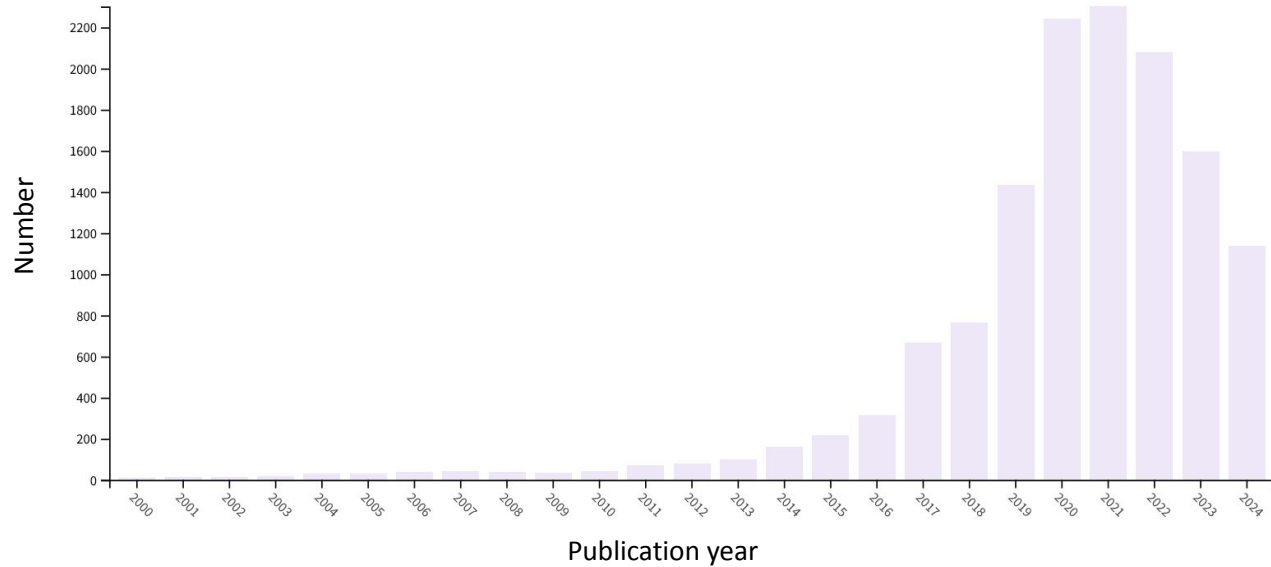
[JOURNAL METRICS >](#)

Online ISSN: 2688-8319

Print ISSN: 2688-8319



Where can a protocol be published?



TI = ((review OR systematic OR meta-analysis)
AND (protocol))

Where can a protocol be published?

Welcome to PROSPERO

International prospective register of systematic reviews

A global registration system for titles and protocols of environmental evidence reviews and syntheses

What is PROCEED?

PROCEED is a global database of prospectively registered evidence reviews and syntheses in the environmental sector. It provides an open access resource of titles and protocols of environmental evidence reviews/syntheses. Authors can register and upload their titles and protocols using appropriate templates. The database is open-access and free to all.



Go to PROCEED

<https://www.proceedevidence.info/>

Where can a protocol be published?



Systematic Review Protocol

Title

What is the influence on socio-economic well-being of UNESCO biosphere reserves in Southeast Asia? A systematic review protocol

Citation:

Phuong Thao Nguyen, Duong Minh Lam, Jacquelyn Eales. What is the influence on socio-economic well-being of UNESCO biosphere reserves in Southeast Asia? A systematic review protocol: a Systematic Review Protocol. PROCEED-22-00029 Available from: <https://www.proceedevidence.info/protocol/view-result?id=29> <https://doi.org/10.57808/proceed.2022.5>

Corresponding author's email address

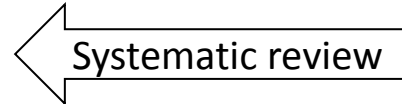
j.f.eales@exeter.ac.uk

Keywords

UNESCO biosphere reserves, conservation, Southeast Asia, human well-being, socio-economics

Background

This PROCEED submission follows the open access a-priori availability of the protocol at Zenodo, prior to commencing this review, on 27th October 2020. DOI: 10.5281/zenodo.4136658 The concept of Biosphere Reserves was introduced in 1975 (Jaisankar, Velmurugan, & Sivaperuman, 2018) by UNESCO in response to the need for conservation of biodiversity along with its sustainable use. Biosphere reserves comprise terrestrial, marine and coastal ecosystems for the purpose of preserving genetic diversity in representative ecosystems by protecting wild animals, the traditional



Systematic Map Protocol

Title

What evidence exists on the potential of Technosols constructed from mineral wastes to host biodiversity?

Citation:

Dakis-Yaoba Ouédraogo, Romain Sordello, Yorick Reyjol, Thomas Lerch. What evidence exists on the potential of Technosols constructed from mineral wastes to host biodiversity?: a Systematic Map Protocol. PROCEED-22-00018 Available from: <https://www.proceedevidence.info/protocol/view-result?id=18> <https://doi.org/10.57808/proceed.2022.3>

Corresponding author's email address

dakis-yaoba.ouedraogo@mnhn.fr

Keywords

Anthroposol; Anthrosol; Circular economy; Constructed Technosol; Ecological engineering; Excavated materials; Urban construction wastes

Background

In 2018, an estimated 55.3 per cent of the world's population lived in urban settlements. By 2030, urban areas are projected to house 60 % of people globally and one in every three people will live in cities with at least half a million inhabitants [1]. The development of cities and transport infrastructures will produce a large volume of excavated materials. For instance, in France, the construction of the Grand Paris Express transport infrastructure will generate 45 million tonnes of these materials. The management of excavated materials, considered as wastes, has a substantial economic and environmental cost (e.g. greenhouse gas emissions), as they are most often stored in

Systematic map



MESSAGES CLÉS

- 1 | Identify the question that matters most
- 2 | Identify and define the different components or concepts
- 3 | Be aware of the problems of not publishing a protocol
- 4 | Choose an appropriate repository aligning with your time and resources